Alexandros Tzanakakis

Machine Learning Engineer · Genomics & Bioinformatics AI

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Education

The University of Texas at Austin — *Austin, TX, USA*
Ph.D. in Electrical and Computer Engineering — *Aug. 2026 – Present*
Research in generative AI, genomic foundation models, and computational biology.

National Technical University of Athens (NTUA) — *Athens, Greece*
M.Eng. in Electrical and Computer Engineering (GPA: 8.4/10) — *Oct. 2018 – June 2025*
Thesis: Segmentation of 3D Medical Images Using Deep Learning.

Technical University of Denmark (DTU) — *Lyngby, Denmark*
Exchange Semester in Computer Science / AI — *Jan. 2022 – May 2022*
Coursework in AI, control system design, and distributed systems.

Skills

Programming	Python, PyTorch, MONAI, Git, Linux, C
Deep Learning	Generative Models (LLMs, gLMs, VAEs), Medical Image Analysis, 3D Segmentation
ML Engineering	Training Pipelines, Model Optimization, Experiment Tracking, Data Processing, Evaluation Metrics
Languages	Greek (Native), English (ECPE, IELTS), French (DELF B2), Japanese (JLPT N4)

Experience

Laboratory of Histology–Embryology, Medical School, University of Crete — *Heraklion, Greece*
Research Intern — *Feb. 2024 – July 2024*
- Hands-on experience in basic cell culture and molecular biology skills (DNA extraction, PCR, Real-time PCR).
- Implementation of cell culture assays (cell growth, adhesion and migration)
- Strengthened scientific communication and interdisciplinary collaboration in a biomedical research setting.

Innovative Compute Core Project, Fujitsu Research — *Tokyo, Japan*
Research Intern & Vulcanus in Japan Participant — *Oct. 2024 – Mar. 2025*
- Designed and implemented a dual-encoder hierarchical VAE combining topological/chemical features with 3D Euclidean NNP embeddings to create fused latent representations for polymers.
- Trained, fused, and evaluated the model end-to-end, integrating Gumbel-Softmax discretization, factorization-machine regression, and Digital Annealer–based QUBO optimization to generate polymers with improved and extrapolated property profiles.

Biomedical Engineering Laboratory, NTUA (ECE) — *Remote / Athens, Greece*
Master's Thesis Researcher — *Feb. 2024 – June 2025*
- Conducted research on 3D brain tumor segmentation (meningiomas) in MRI using classical CNNs, transformer-based, and hybrid deep learning architectures.
- Designed a unified experimental pipeline to compare models under realistic constraints of limited data and compute.
- Optimized preprocessing, augmentation, and evaluation workflows for reproducible medical imaging experiments.

Georgakopoulos–Soares Lab (Penn State → UT Austin) — *Remote / Austin, TX*
Research Scientist (AI for Genomics) — *Apr. 2025 – Present*
- Analyzed DNA k-mer frequency patterns across >225,000 genomes, one of the largest datasets used for this purpose, demonstrating how genome size and composition drive vocabulary expansion via Heaps' law and heavy-tailed models.
- Developed diagnostic frameworks comparing real genomes to sequences produced by genomic LLMs, identifying systematic deviations in k-mer spectra, Non-B DNA motifs, CpG islands, TFBS frequencies, and repeat content.
- Conducted a comprehensive analysis of k-mer symmetry classes across >200,000 genomes using Markov models to separate routine compositional bias from true strand-, replication-, and structure-related asymmetries.
- Contributed to the development of deterministic speculative decoding algorithms for genomic foundation models, creating fast inference strategies that yield 2–$5\times$ faster genomic sequence generation.

Honors & Awards

2026–27 **$6,000/year Fellowship**, Cockrell School of Engineering & UT Austin Graduate School — *Austin, TX*

2026 **ACCESS Discover Compute Allocation**, NSF ACCESS Program – 750,000 ACCESS Credits for PhageForge — *United States*

2016 **6th Prize**, Nationwide Chemistry Competition (2nd Grade High School) — *Greece*

2017 **Top 0.0005% Nationwide**, Panhellenic Exams (University Entrance Assessment) — *Greece*

Conference Presentations

Repeated-run Reliability of Biomedical Agents: Measurement Pitfalls and Selective Deferral

NeurIPS 2026 Workshop on Agentic Learning Systems (AgenticLS)

Poster — *2026*

- Tzanakakis A., Georgakopoulos-Soares I.
- Accepted for poster presentation at NeurIPS 2026.

A Hierarchical Variational Autoencoder for Polymer Design using Fused Latent Representations

CECAM Flagship Workshop on Virtual Materials Design, KIT (Germany)

Poster Presenter — *Jun. 2024*

- Tzanakakis A., Haderbache A., Yoshimoto Y., Dang T., Oyama Y., Sakai Y.

Publication List

Structure is not mechanism: high-gain gated-FFN rows across text and genomic foundation models

arXiv

First author — *2026*

- Tzanakakis A., Karatzikos A., Georgakopoulos-Soares I.
- `arXiv:2609.17599`

Limitations of Genomic Language Models for Realistic Sequence Generation

Patterns

First author — *2026*

- Tzanakakis A., Mouratidis I., Georgakopoulos-Soares I.
- doi: `10.1016/j.patter.2026.101682` ISSN: 2666-3899

Calibration-free compression brings Evo 2 to its full million-token context on a single GPU

bioRxiv

Co-first author — *2026*

- Patsakis M., Tzanakakis A., Georgakopoulos-Soares I.
- Status: Preprint

Accelerating Inference in Genomic and Proteomic Foundation Models via Speculative Decoding

Bioinformatics

Co-author — *2026*

- Provatas K., Karatzikos A., Koilakos C., Patsakis M., Tzanakakis A., Nayak A., Pavlopoulos G.A., Mouratidis I., Avgoulas E.I., Georgakopoulos-Soares I.
- doi: `10.1093/bioinformatics/btag579`

A Padding Method for Enhanced Encoding of Inorganic Structures with Varying Chemical Compositions

arXiv

Co-author — *2026*

- Dang T., Amir H., Tzanakakis A., Yoshimoto Y.
- `arXiv:2605.30743`

Investigating DNA Words and Their Distributions Across the Tree of Life

Computational and Structural Biotechnology Journal

Co-author — *2025*

- Koilakos C., Provatas K., Patsakis M., Karatzikos A., Tzanakakis A., Georgakopoulos-Soares I.
- doi: `10.1016/j.csbj.2025.11.040` ISSN: 2001-0370

Enhancing Tumor Targeted Therapy: The Role of iRGD Peptide in Advanced Drug Delivery Systems

Cancers (Basel)

Co-author

2024

- Nikitovic D., Kukovyakina E., Berdiaki A., Tzanakakis A., Luss A., Vlaskina E., Yagolovich A., Tsatsakis A., Kuskov A.
- doi: `10.3390/cancers16223768` PMID: 39594723 PMCID: PMC11592346